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COWI FONDEN

FLOATING SOLAR PV RESEARCH FACILITY

REQUEST FOR PROPOSAL

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1 Invitation and Purpose

DTU and COWI invite you to submit a detailed proposal for the engineering, supply, installation, testing and commissioning of an approximately 30-40 kWp floating solar photovoltaic (FPV) installation in Denmark. The intended contracting model is a complete turnkey delivery, with the supplier responsible for all works, equipment and coordination required to provide a safe, compliant and fully operational plant, except for items expressly identified as supplied by DTU/COWI or excluded in the proposal.

The offer should be sufficiently transparent to allow technical and commercial comparison between the proposed solutions. In particular, please provide separate prices for the principal work packages and equipment categories identified in Section 4.4 Financial proposal and price breakdown.

An optional quotation may be submitted for an additional floating system, provided that it presents sufficiently different technical characteristics and specifications.

2 Project Overview

This innovative research project seeks to establish and operate an experimental floating solar PV test facility at the Photonic Lab at DTU in Risø, Roskilde.

The project will investigate, analyse, and discuss the technical challenges associated with offshore floating PV system design and evaluate the market and performance potential of floating solar energy in offshore environments in Denmark and elsewhere. Furthermore, the project will evaluate the impact of floating photovoltaic systems on the flora and fauna in the surrounding area.

A series of configurations will be tested to investigate its individual profitability, reduce the risks and uncertainties regarding floating solar projects for public and private investors, and generate reliable data to assist in developing standards, regulations, and certifications for the floating solar market.

After the project period has ended, the results, certain data and reports will be publicly available.

This pilot project consisting of different floating PV systems will be setup, tested and compared based on a series of parameters including performance, LCOE, O&M, marine life impact, and more.

Table 1 Project Overview

Parameter	Tender basis
Country	Denmark
Plant type	Floating solar photovoltaic research installation
Indicative DC capacity	Approximately 30-40 kWp
Final installed capacity	To be confirmed by the detailed design and accepted by DTU/COWI
Delivery Scope	Engineering, procurement/supply, installation, commissioning and handover
Grid / point of connection	TBD
Site data	Section 2.1 Please identify any site data, and surveys required to finalize the proposed design and pricing
Start construction	Estimated construction start: April the 5th 2027
COD	Estimated construction end: May the 3rd 2027

Where information is not available, explicit assumptions, qualifications and data requirements must be stated. Any material dependency on third parties, permits, grid conditions, water level variation or DTU/COWI inputs shall be clearly identified.

2.1 The research facility

The different floating solar systems will be selected and installed west of the Risø campus in Roskilde Fjord, which is located close to DTU Electros Photovoltaic research facility, where there is a small harbour place as shown in the following figures.

Permission for these areas has been seeded. The water is shallow <4 m in the harbour area but rises to >5m in the light blue area where a floating bridge will be extended, and the testbed placed.

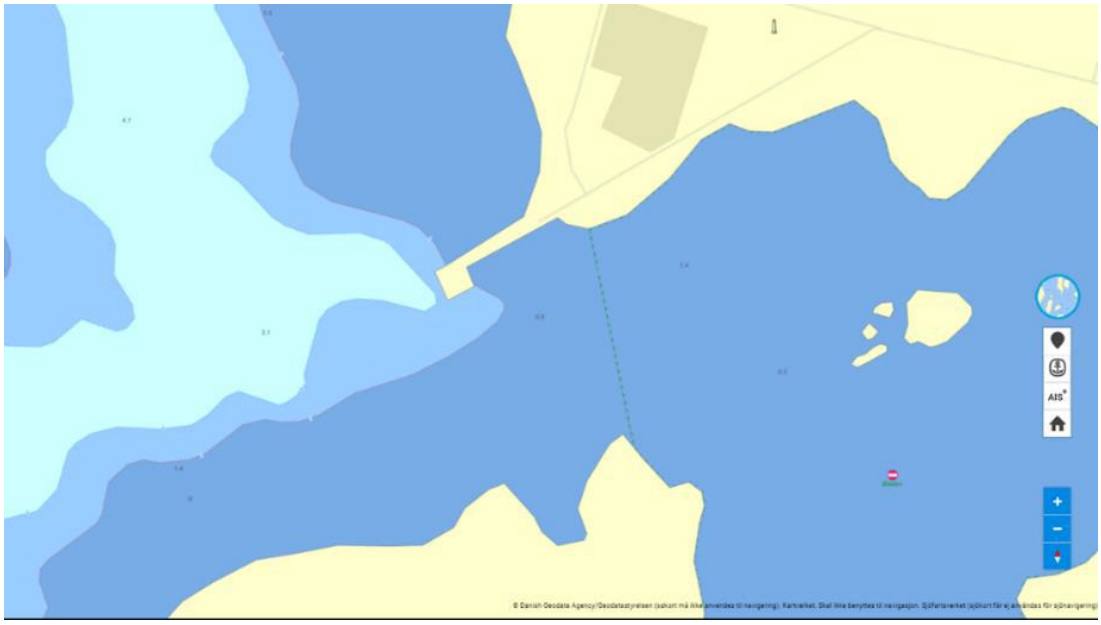


Figure 1 Illustration of water depths near Risø Campus. Lighter colours mean deeper waters.

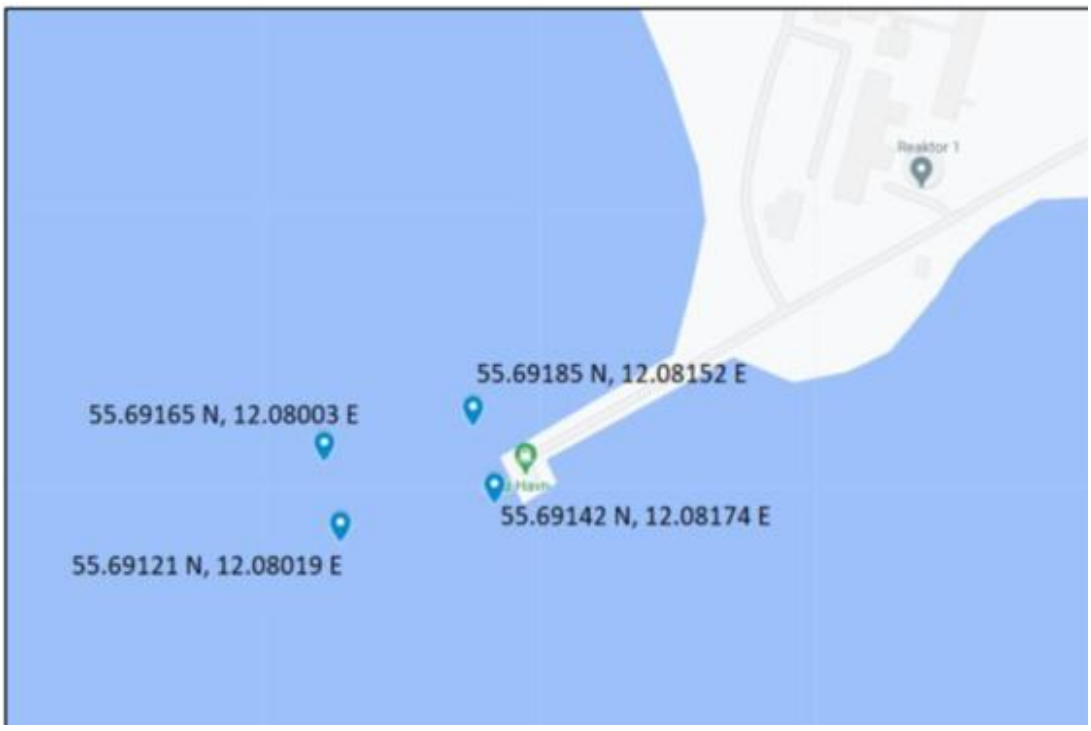


Figure 2 Project location (©Google Maps).

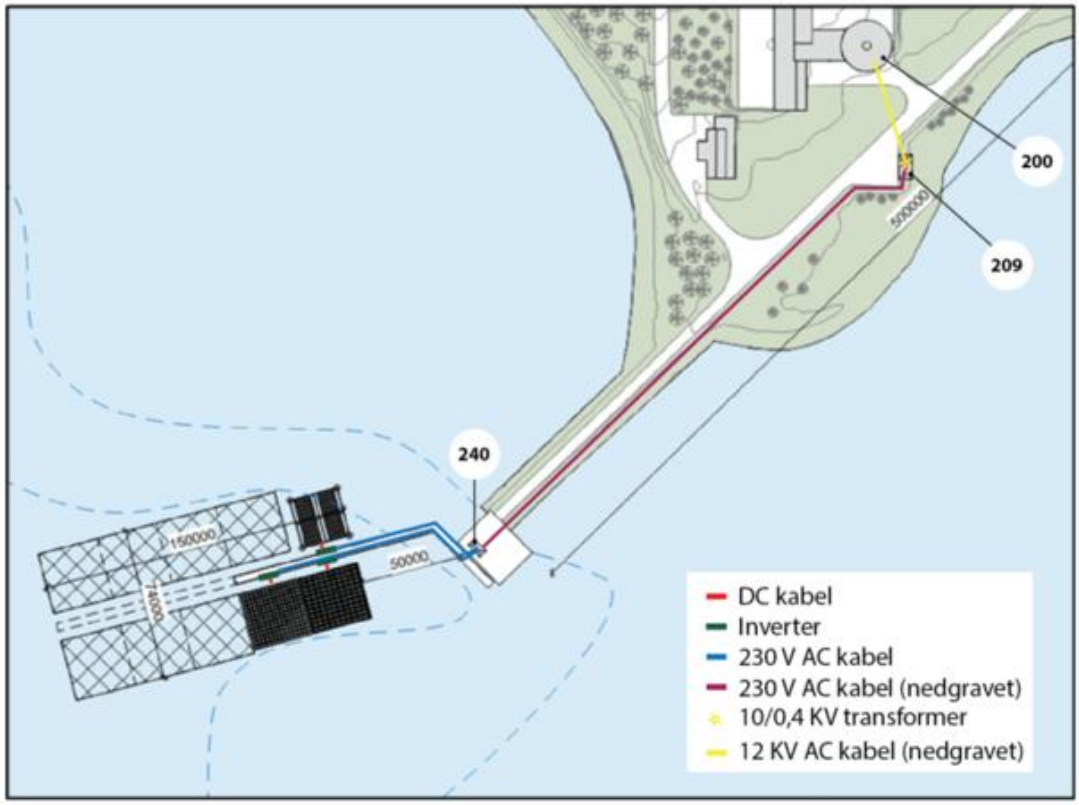


Figure 3 Illustrative potential solar floating systems.

Several other ground-mounted PV projects along with a wind power project are currently running at the facility, where detailed solar resource monitoring data is available as well as a good research infrastructure, allowing relatively easy data collection.

3 Tender Basis

3.1 Scope of Supply and Services

The suppliers will deliver the floating systems, mooring and anchoring, PV modules, complete DC installations and inverters mounted on the pontoons/floating bridge, as illustrated in Figure 3, including earthing and bonding.

AC cabling, AC switchgear, any transformer and the associated shore-side electrical works, as well as additional research sensors are handled separately on the DTU/COWI side.

The proposed scope is to include the following (but not limited to) activities unless expressly excluded and priced as an option.

3.1.1 Engineering and Design

- Project-specific design basis and design criteria, including applicable loads, environmental conditions, water-level range and operating conditions.
- PV layout and electrical design, string configuration, inverter sizing, cable sizing, voltage-drop calculations and electrical single-line diagrams.
- Floating platform layout and structural verification, including module-to-float interfaces, walkways/access provisions and freeboard as applicable.
- Mooring and anchoring design appropriate to the specific water body, including design loads, anchor type, line configuration, bank/bottom interfaces and allowance for water-level variation.
- Protection, earthing/bonding, surge/lightning protection, metering interfaces and connection up to the agreed limit of the scope of work.
- Cable routing and management for floating sections, including dynamic/flexible cable provisions where required.
- Monitoring, plant controller / communications and SCADA interface design, where applicable.
- Safe isolation, electrical safety and O&M practices.
- Design calculations, drawings, equipment schedules, datasheets, and method statements.

3.1.2 Equipment Supply

- PV modules, including product/performance warranties.
- Floating system, pontoons/floats, structural elements, walkways, connection hardware and module mounting accessories.
- Mooring and anchoring system, including anchors, ropes/chains/lines, shackles, connectors, tensioning components and shore interfaces.
- Inverters and associated mounting/support arrangements.
- DC cables, connectors, string/combiner equipment where required, labels and cable management components and all auxiliary electrical DC equipment within the supplier scope.
- Monitoring, communications and SCADA equipment included in the proposed solution, if any.
- All consumables, fastening systems, fixings, glands, conduits, trays and ancillary materials needed for a complete installation.
- Spare parts and special tools proposed for initial operation and maintenance.

3.1.3 Installation and Construction

- Mobilisation, site establishment, temporary works, and demobilisation.
- Assembly and installation of the floating array and PV modules.

- Installation and proof/verification of anchors and mooring system in accordance with the accepted design.
- Installation of DC, earthing/bonding, communications and shore-side interfaces.
- All civil works within the tendered scope.
- Quality control, inspection and construction records.
- Housekeeping and reinstatement of areas affected by the Works.

3.1.4 Testing, Commissioning and Handover

- Pre-commissioning inspections and electrical tests.
- Functional testing of inverters, protection, monitoring, communications and safety systems.
- Commissioning of the complete FPV plant and demonstration of correct operation at the agreed point of connection.
- PV string/array testing and test records appropriate to the supplied system.
- Mooring/anchoring completion records and final inspection.
- Punch-list.
- O&M manuals if any, warranties, certificates, test reports and commissioning records.

3.2 Floating PV Specific Design Requirements

- The floating solar system shall be designed and constructed to a high standard.
- The proposed floating technology should be a proven commercial solution suitable for the project water body and project climatic and environmental conditions (wind, wave, current, temperature, etc).
- The design life of the floats, structural components, mooring lines and anchoring system should be stated together with any associated warranties.
- Materials exposed to water, humidity or splash shall be suitable for the expected environment and selected to control corrosion, degradation and galvanic interaction.
- Safe access to the array for inspection, maintenance and emergency response should be described.
- Any restrictions or considerations related to the Natura 2000 location should be identified relevant to the technology/materials.

3.3 Codes, Standards and Regulatory Compliance

The works shall comply with all applicable Danish legislation and Standards, authority requirements, grid-connection requirements and the latest applicable editions of relevant European and international standards.

3.4 Health, Safety, Environment and Quality

The works shall comply with applicable Danish occupational health and safety requirements and site-specific DTU/COWI rules.

Rigorous quality and safety protocols shall be maintained throughout the project. This includes:

- Relevant certifications such as ISO 9001 (Quality Management) and ISO 45001 (Occupational Health and Safety).
- Consistent quality control and adherence to all safety regulations.

The project HSE approach shall cover work near/on water, electrical works, lifting operations, workboats, rescue arrangements and emergency response.

Prior to construction activities, the following information should be provided:

- Relevant method statements and risk assessments.

- Description of proposed environmental controls for spills, waste, materials entering the water body, noise and disturbance during installation.
- Quality plan / inspection and test plan.

3.5 Information by DTU and COWI

DTU/COWI will manage the following project permits and consents:

- Facility permit
- Grid connection

The grid connection to DTU facilities (ref. Figure 3) will be handled by DTU Campus service.

The following information will be confirmed by DTU/COWI before contract:

- Grid connection point, voltage level, available capacity and DTU technical conditions.
- Existing electrical infrastructure to be reused, if any.
- Permitting status and responsibilities (such as land building permit, etc.).
- Construction access, laydown and launch areas.
- Target dates and any operational restrictions on the water body.
- DTU/COWI contracting requirements and final form of contract.
- DTU specific H&S requirements.

3.6 Tender Programme and Submission

Table 2 Timeline

Milestone	Date
RFP issued	October 23, 2026
Deadline for questions	October 30, 2026
Proposal submission deadline	November 6, 2026
Preferred date / signature of Agreement	December 2026
COD date	May 2027

3.6.1 Submission Format:

The proposal shall be submitted electronically in English and shall remain valid for the period stated on the cover page. Commercial pricing should be provided in EUR, exclusive of VAT.

All material, commercial and technical documents are sent to:

Morten Frøkjær Petersen: mfpn@cowi.com

Emma García: emga@cowi.com

Martin Bartholomäus: mabart@dtu.dk

Peter Behrensdorff Poulsen: ppou@dtu.dk

Contact for queries: Please direct all questions regarding the tender material to the contacts above.

3.7 Evaluation of Proposals

Price – 85%

The evaluation will be based on the evaluated tender price. The evaluation may take into account technical suitability of the floating and electrical solution, proposed schedule, assumptions, exclusions and overall project value.

Technical solution and Timeline – 15%

The evaluation will be based on the technical solution and compliance with DTU/COWI requirements, schedules, permits and project-specific conditions.

3.8 Confidentiality and Compliance

Information provided in this Request for Proposal (RFP) and any subsequent documentation is confidential and should not be disclosed without prior consent.

4 Information to Be Submitted

The following information must be included in the proposed solution:

4.1 Technical proposal

- Description of the proposed FPV solution and key design assumptions.
- Preliminary layout showing floating array arrangement, inverter location(s), mooring concept, cable routes and inverter connection.
- Equipment schedule with manufacturer, model, quantity and key ratings for modules, floats, inverters, mooring/anchoring, and monitoring equipment.
- PV module, inverter and floating-system datasheets.
- Description of the mooring/anchoring concept and required site data for final design.
- Preliminary electrical single-line diagram.
- Statement of the precise electrical boundary of the proposal and all assumptions regarding the existing installation, connection voltage, meter, protection settings and DTU interface.
- Energy-yield estimate or expected annual production, with assumptions and loss factors clearly stated.
- Outline construction and commissioning methodology.
- List of subcontractors and key suppliers proposed, if any.
- Product warranties for key equipment and performance warranties for the facility.
- Please describe any proposed plant availability or performance guarantee together with assumptions / exclusions applicable to it.
- Please provide recommended preventive maintenance activities and, if available, an optional price for annual O&M support.
- Indication of which surveys and investigations are assumed available, and which are expected to be carried out as part of the proposed solution (bathymetry/topography, geotechnical/anchor testing, water level and utility detection).
- Any notifications, permits, approvals or inspections considered necessary should be identified together with an indication of whether these are included in the proposal.
- Identify any dependencies on DTU, COWI or third parties that could affect the proposed scope, schedule or pricing.
- Statement of requirements for access, workboats, cranes, launch areas, temporary storage and any restriction affecting price or programme.
- Statement of waste quantities and the practical arrangements required for waste management.

4.2 Commercial proposal

- Start date and a binding or indicative delivery programme, clearly identified as such. Project programme shall identify engineering, procurement, manufacturing, installation, commissioning and handover milestones.
- Identify procurement constraints, if applicable
- Summary of the main commercial and contractual assumptions, including:
    - Statement of quotation validity, payment milestones, Incoterms where relevant, taxes/VAT treatment and currency.
    - Statement of any advance payment or other security required, if applicable.
    - Description of any proposed limitations, assumptions or commercial qualifications relevant to the proposal.

- Relevant experience with floating PV projects of comparable nature, with references and contact details where permitted.

4.3 Reservation list & Deviations

All clarifications, deviations, exclusions, assumptions and optional items shall be consolidated in a single schedule in the proposal.

If no deviations are identified, the proposal will be assumed to generally comply with the requirements described in this document.

4.4 Financial proposal and price breakdown

Prices shall include all costs necessary to deliver the stated scope, except items expressly excluded.

Please provide prices separately. Sub-items may be added where this improves transparency.

Prices should be provided in EUR, excl. VAT. To facilitate management of scope adjustments, a 35 kWp system should be considered. For each of the items below, please also provide EUR/kWp rates.

Table 3 Mandatory price breakdown

Item	Price (EUR)	Unit Price (EUR/kWp)
1 Engineering, project management, transport and logistics, site mobilisation and temporary work		
2 PV modules		
3 Floating platform / mounting system		
4 Mooring and anchoring system		
5 Inverter		
6 Electrical balance of system		
7 Monitoring / communications / SCADA		
8 Mechanical and Electrical installation		
9 Civil / shore-side works		
10 Testing and commissioning, documentation and training		
11 Initial spare parts and special tools		
12 End-of-life removal and disposal costs		
13 Other costs		
14 TOTAL TURNKEY PRICE		

4.4.1 Options to be priced separately

Table 4 Optional price

Item	Price (EUR)	Unit Price (EUR/kWp)
15 Bridge / Pontoon for access to all floating systems		-
16 Total price excluding inverter procurement and supply		
17 Discount for proposals including two different floating systems (of 35 kWp each)		